Wheely wiring reference

Raspberry Pi Pico 2 W + RCWL-1601 ultrasonic + Adafruit DRV8833. Bench configuration: Pico powered from USB, battery powers the motor driver only. Pin numbers verified against the official Pico 2 W pinout (datasheets.raspberrypi.com). Generated 2026-10-06.

1. Pico 2 W pins used by this project

Pin	Pico label	Connects to	Role / note
1	GP0	DRV8833 AIN1	PWM slice 0 ch A. Motor A direction/speed
2	GP1	DRV8833 AIN2	PWM slice 0 ch B. Motor A direction/speed
3	GND	Breadboard ground rail	Common ground for sensor, driver and battery –
21	GP16	RCWL-1601 Trig	Output. 10 µs pulse starts a ping
22	GP17	RCWL-1601 Echo	Input, pull-down. High for the round-trip time
36	3V3(OUT)	Breadboard 3.3 V rail	Feeds sensor VCC and DRV8833 SLP
4	GP2	DRV8833 BIN1 (later)	PWM slice 1 ch A. Motor B, milestone 3+
5	GP3	DRV8833 BIN2 (later)	PWM slice 1 ch B. Motor B, milestone 3+
39	VSYS	Battery + via Schottky (later)	Only when the robot runs untethered. Not now
30	RUN	Optional reset button to GND	Hold BOOTSEL + tap RUN = BOOTSEL without unplugging

Green = wired now. Yellow = reserved for later. Never connect battery + to 3V3(OUT) or VBUS. The onboard LED is on the Wi-Fi chip (WL_GPIO0), not a header pin.

2. Full Pico 2 W header map (USB connector at the top)

Pin	Label	Used for		Pin	Label	Used for
1	GP0	AIN1		40	VBUS	USB 5 V. Leave alone
2	GP1	AIN2		39	VSYS	battery + later (diode)
3	GND	ground rail		38	GND	
4	GP2	BIN1 later		37	3V3_EN	
5	GP3	BIN2 later		36	3V3(OUT)	3.3 V rail
6	GP4			35	ADC_VREF	
7	GP5			34	GP28 / ADC2	
8	GND			33	GND / AGND	
9	GP6			32	GP27 / ADC1	
10	GP7			31	GP26 / ADC0	
11	GP8			30	RUN	reset, optional
12	GP9			29	GP22	
13	GND			28	GND	
14	GP10			27	GP21	
15	GP11			26	GP20	
16	GP12			25	GP19	
17	GP13			24	GP18	
18	GND			23	GND	
19	GP14			22	GP17	Echo
20	GP15			21	GP16	Trig

DEBUG connector (3-pin JST-SH at the bottom edge): SWCLK, GND, SWDIO. For the Debug Probe later; unused now.

3. RCWL-1601 ultrasonic sensor (3.3 V native, no level shifter)

Sensor pin	Goes to	Note
VCC	3.3 V rail (from Pico pin 36)	Do not use VBUS 5 V; the Echo pin would then be 5 V into the Pico
Trig	GP16 (pin 21)	
Echo	GP17 (pin 22)	Firmware enables the pull-down so an unplugged Echo reads as NoEcho
GND	ground rail	

4. Adafruit DRV8833 breakout

Driver pin	Goes to	Note
VM	Battery +	Motor supply, 2.7 to 10.8 V. 4xAA = 6 V alkaline / 4.8 V NiMH
GND	ground rail AND battery –	All grounds on the breakout are connected
SLP	3.3 V rail	Internal 500 kΩ pull-down: chip is ASLEEP unless this is high
FLT	not connected	Open-drain fault output. Could go to a GPIO later
AIN1	GP0 (pin 1)	Motor A
AIN2	GP1 (pin 2)	Motor A
AOUT1 / AOUT2	Motor A wires	Swap the two wires if 'forward' runs backwards
BIN1 / BIN2	GP2 / GP3 later	Motor B, not yet
BOUT1 / BOUT2	Motor B later	

Input pair truth table (same for A and B):

IN1	IN2	Motor
PWM	0	forward at duty
0	PWM	reverse at duty
0	0	coast
1	1	brake (unused)

5. Power rules for the bench

• Pico on USB (micro-USB to the Mac). Battery + goes to DRV8833 VM only. Battery – goes to the ground rail.
• One jumper from Pico pin 36 to the red rail; one from pin 3 to the blue rail. Everything else hangs off the rails.
• Battery switch OFF while wiring and while flashing. Switch ON only after the serial log is visible.
• Later, untethered: battery + to VSYS through a Schottky diode; add 100 to 470 µF across the battery leads near the driver if the Pico resets when motors start.

6. Flash and watch (no Debug Probe)

```
# hold BOOTSEL while plugging in the Pico, then:
cd ~/projects/wheely
cargo run --release --bin sonar # ranging test (or just: cargo run --release)
cargo run --release --bin motor # DRV8833 motor A ramp test
screen /dev/cu.usbmodem* 115200 # logs; quit with Ctrl-A, K, y
```

Sources: Pico 2 W pinout, datasheets.raspberrypi.com/picow/pico-2-w-pinout.pdf. DRV8833 pin behaviour, learn.adafruit.com/adafruit-drv8833-dc-stepper-motor-driver-breakout-board/pinouts. Project code: ~/projects/wheely/src/bin/{sonar,motor}.rs.